



PRASHANT SINGH

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Ph.D. researcher in Computational Biology at IIT Hyderabad, specializing in automated genome-scale metabolic model reconstruction using machine learning and graph-based approaches. Experienced in developing bioinformatics pipelines for genomics, drug repurposing, and antimicrobial resistance analysis. Proficient in Python, R, and systems biology tools, with a growing interest in applying NLP and large language models for biomedical knowledge extraction and pathway curation.

Committed to advancing computational methods at the intersection of systems biology, AI, and precision medicine.

EDUCATION

Ph.D. in Computational Biology, IIT Hyderabad

Aug 2023- Present

Thesis Title: Automated Reconstruction of Genome-Scale Metabolic Models for Micro-eukaryotic Parasites.

Brief Summary: Developing ML-driven pipelines integrating graph-based algorithms and multi-omics data to automate model reconstruction, enabling high-throughput phenotype prediction and biomarker discovery for neglected parasitic diseases.

M.Sc. Bioinformatics, Maharshi Dayanand University (Rohtak)

Sep 2020

Thesis Title: Breast cancer prediction modelling using machine learning approaches and model comparisons.

B.Sc. Biotechnology, Pt. J L N Govt College (Faridabad)

May 2017

Senior Secondary, HBSE, (Bhiwani)

June 2014

Sub: PCMB

COMPUTER SKILLS

- Operating Systems: Windows, Unix, Linux, high-performance computing (HPC) environments
- Programming Skills: Python, R, MATLAB, HTML, Perl, JavaScript
- Application and Tools: MEGA, PHYLIP, Bioedit, ChemSketch, MARVIN, UCSF chimera, Rasmol, SPDV, Pymol, AUTODOCK-Vina, PyRx, Discovery-studio, GROMACS, SQL, Prokka, RAST, MODELLER, CobraToolbox
- Bioinformatics Skills: Machine learning, Computational analysis, Data Visualization, Genomics

RESEARCH EXPERIENCE

Project Associate, Centre for Computational Biology and Mathematical Modelling, THSTI

June 2022 – August 2023

- Developed “MegaResistome” pipeline and their graphical user interfaces (GUI) for ARGs prediction and analysis from WGS data. These were developed in Python FLASK.
- Developed a tool for identification of antibiotic induced regulatory elements in the genome of the MDR pathogens.

Project Technical Staff, Bioinformatics and Data Management, National Institute of Virology

Feb 2022 – April 2022

- Worked in Covid data analysis research program runs by ICMR and Govt. of India. In that project I have tasked with analysis of WGS data collected from different VRDL labs and hospital across country.
- Here our duty was to find out the strain information of the sequencing data and prepare it for GISAID submission.

Project Associate, Amity Institute of Biotechnology, Amity University, Noida

Feb 2021 – Jan 2022

- Developed AI based pipeline and their graphical user interfaces (GUI) for repurposing of Drugs for SARS-CoV-19. These were developed in Python FLASK.
- Contributed to reverse vaccine development project for Chagas Disease, funded by BMC, U.S.

SEMINARS ATTENDED

- Attended 5th IBSE International Symposium, IIT Madras, Chennai, Feb 19-21, 2025
- Computational workshop on “Analysis of Next-Generation Sequencing Datasets from Bulk, Single-Cell, and Spatial Transcriptomics”, ICGEB, Delhi, Aug 7- 9, 2024
- International Meet on Preparedness for future Epidemics, CEPI, THSTI, Faridabad, Dec. 5-6, 2022
- Computational workshop on Genomics, Proteomics and Metagenomics-2022, IGIB, July 20-23, 2022

PUBLICATIONS AND CONFERENCE PRESENTATIONS

International Peer Reviewed

- Brain peptides in Alzheimer’s disease-pathophysiology and therapeutic advances, Cell Tissue Res., 10.1007/s00441-026-04055-8
- Trehalose metabolism regulates transcriptional control of muscle development in lepidopteran insects, Elife. 10.7554/eLife.109485.1
- Identification of vaccine targets in pathogens and design of a vaccine using computational approaches, Nature Sci. Rep. PMID: 34475453

Preprints

- Singh, Prashant, Robin Sinha, Swarsat K. Nath, Preeti P, Priya Kumari, and Nikita garg. 2021. “COV-DOCK Server: A Web Server for COVID-19 Ligand-target Docking.” OSF Preprints. November 29. doi:10.31219/osf.io/x2ky5
- Rawal, Kamal, Prashant Singh, Robin Sinha, Priya Kumari, Swarsat K. Nath, Nikita garg, Preeti P, et al. 2021. “Cov-drug Target Interaction Server for Covid-19 Drug Repurposing.” OSF Preprints. December 7. doi:10.31219/osf.io/gejwq
- Pendharkar, R. A., Preeti, P., Singh, P., Sinha, R., Kumari, P., Nath, S. K., ... & Rawal, K. (2022). Analysis of Indomethacin as a potential drug for COVID-19 using CoV-DrugX Pipeline.
- Verma, N., Singh, P., Sinha, R., Preeti, P., Sharma, T., Nath, S. K., & Rawal, K. (2022). Impact of Quercetin on Sars-cov-2 Using Drugx Repurposing Pipeline.

Conference Presentation

- A web application for comprehensive ARG prediction from WGS data using multiple resistome databases, HMCW2023, THSTI, Faridabad. Feb. 15-17, 2023

RESEARCH INTERESTS

- Applying NLP and large language models (LLMs) to automate biological knowledge extraction, pathway curation, and hypothesis generation for systems-level disease modeling — directly aligned with current advances in AI-driven biomedical discovery.
- Automating genome-scale metabolic model reconstruction using graph-based ML, enabling high-throughput phenotype prediction and biomarker discovery from multi-omics data.
- Leveraging computational systems biology to decode the genomic landscape of cancer and infectious diseases, identifying novel therapeutic targets through integrative multi-omics analysis.
- Combating antibiotic resistance through the application of AI in the discovery and development of new antimicrobial agents

HONORS/AWARDS

- Got 1st position in university level slogan writing competition organized by Environment sustainability Management cell, Maharshi Dayanand University, Rohtak.
- Organized Bio-info Fest 2019 organized by CBINF, MDU, Rohtak
- Got 2nd position in DISTRICT LEVEL SCIENCE QUIZ COMPETITION Sponsored by State council for Science and Technology, Haryana, Chandigarh.
- Participated in Zonal Level Science Quiz Contest (school) organized by Haryana State council For Science & Technology, Panchkula.

LANGUAGE SKILLS

Fluent in English and Hindi.

REFEREE LIST

Dr. Abhishek Subramanian

Assistant Professor

Department of Biotechnology and Bioinformatics

Indian Institute of Technology Hyderabad, Sangareddy

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Dr. Sara Cherian

Scientist G

Department of Bioinformatics and Data Management

ICMR-National Institute of Virology, Pune

cherian.ss@gov.in

Dr. Amit Kumar Yadav

Principal Investigator - II

Centre For Computational Biology and Mathematical Modelling

DBT - Translational Health Science and Technology Institute, Faridabad

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